

Lab (Option B: JavaScript): Build and Deploy a Simple Guestbook App



Objectives

In this lab, you will:

- Build and deploy a simple Guestbook application
- Autoscale the Guestbook application using Horizontal Pod Autoscaler
- Perform Rolling Updates and Rollbacks

Project Overview

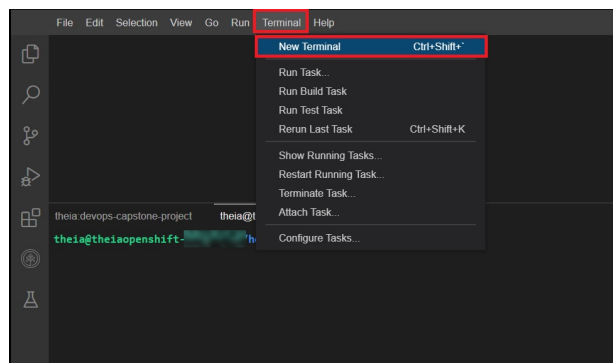
Guestbook application

Guestbook is a simple web application that you will build and deploy with Docker and Kubernetes. The application consists of a web front end which will have a text input where you can enter any text and submit. For all of these, you will create Kubernetes Deployments and Pods. Then, you will apply Horizontal Pod Scaling to the Guestbook application and finally work on Rolling Updates and Rollbacks.

Verify the environment and command line tools

1. If a terminal is not already open, open a terminal window by using the menu in the editor: `Terminal > New Terminal`.

Note: Please wait for some time for the terminal prompt to appear.



2. Change to your project folder.

Note: If you are already on the `/home/project` folder, please skip this step.

```
cd /home/project
```

3. Clone the git repository that contains the artifacts needed for this lab.

```
[ ! -d 'guestbook' ] && git clone https://github.com/ibm-developer-skills-network/fqdrq-guestbook-js.git guestbook
```

```
theia@theiaopenshift-harshsingh15:/home/project$ [ ! -d 'guestbook' ] && git clone https://github.com/ibm-developer-skills-network/fqdrq-guestbook-js.git guestbook
Cloning into 'guestbook'...
remote: Enumerating objects: 15, done.
remote: Counting objects: 100% (15/15), done.
remote: Compressing objects: 100% (14/14), done.
remote: Total 15 (delta 0), reused 14 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (15/15), 24.33 KiB | 12.17 MiB/s, done.
```

4. Change to the directory for this lab.

```
cd guestbook
```

5. List the contents of this directory to see the artifacts for this lab.

```
ls
```

Creating the Guestbook app

Note:

1. Please ensure that all updates to the files are properly saved.
2. Download your project files and capture screenshots as mentioned in the respective tasks.
3. As mentioned in the “*Final Project Overview and Review Criteria*”, you can submit your project deliverables through either Option 1: AI-Graded Submission and Evaluation or Option 2: Peer-Graded Submission and Evaluation.
4. **For Option 1: AI-Graded Submission and Evaluation:**
 - Submission requires the terminal output and code snippet.
5. **For Option 2: Peer-Graded Submission and Evaluation:**
 - Submission requires screenshots for **Tasks 1 to 10**

Build the guestbook app

To begin, build and deploy the web front end for the guestbook app.

1. Change to the `vi/guestbook` directory.

```
cd v1/guestbook
```

2. Dockerfile incorporates a more advanced strategy called multi-stage builds, so feel free to read more about that [here](#).

The Dockerfile with the necessary Docker commands to build and push your image is provided. The path to this file is `guestbook/v1/guestbook/Dockerfile`.

► Hint!

The FROM instruction initializes a new build stage and specifies the base image that subsequent instructions will build upon.

The COPY command enables us to copy files to our image.

The ADD command is used to copy files/directories into a Docker image.

The RUN instruction executes commands.

The EXPOSE instruction exposes a particular port with a specified protocol inside a Docker Container.

The CMD instruction provides a default for executing a container, or in other words, an executable that should run in your container.

Assessment:

For Option 1: AI-Graded Submission and Evaluation: Download or copy and paste the Dockerfile and save it in a text file named `Dockerfile` for the final project submission and evaluation.

For Option 2: Peer-Graded Submission and Evaluation: Take a screenshot of the completed dockerfile and save it as `Dockerfile.jpeg` OR `Dockerfile.png` for the Peer Assignment.

3. Export your namespace as an environment variable so that it can be used in subsequent commands.

```
export MY_NAMESPACE=sn-labs-$USERNAME
```

```
theia@theiaopenshift: /home/project/guestbook/v1/guestbook$ export MY_NAMESPACE=sn-labs-$USERNAME
theia@theiaopenshift: /home/project/guestbook/v1/guestbook$
```

4. Run the following command before building the Docker image.

```
npm install
```

5. Build the guestbook app using the Docker Build command.

► Hint!

```
docker build -t us.icr.io/sny/MY_NAMESPACE/guestbook:v1
```

```
=> [internal] load build definition from Dockerfile                                0.0s
=> => transferring dockerfile: 976B                                              0.0s
=> [internal] load metadata for docker.io/library/node:20-alpine                0.3s
=> [internal] load .dockerignore                                                 0.0s
=> => transferring context: 186B                                                  0.0s
=> [builder 1/4] FROM docker.io/library/node:20-alpine@sha256:effc1ee6b93c4db0aab5729f3bc25d9d 5.4s
=> => resolve docker.io/library/node:20-alpine@sha256:effc1ee6b93c4db0aab5729f3bc25d9df841e9b8 0.0s
=> => sha256:effc1ee6b93c4db0aab5729f3bc25d9df841e9b88ec3c4683d5a948a37553244 1.72kB / 1.72kB 0.0s
=> => sha256:2215267afb33d93392d8d16b41c461a569bf52a442e5af4c4add1371b73f26e3 6.18kB / 6.18kB 0.0s
=> => sha256:38a8310d387e375e0ec6fabe047a9149e8eb214073db9f461fee6251fd936a75 3.64MB / 3.64MB 0.9s
=> => sha256:65052e355180ad17acd4b61613338090c587b931124cd2946b9dcf0e945b76c 42.54MB / 42.54MB 3.1s
=> => sha256:9cf3157062e6e87bb84921361434e0c930e8f73584eb06114cf7563178355253 1.26MB / 1.26MB 0.8s
=> => sha256:dcfb660a4e70b11e73b64689666db5db05706d6d6a9bb950d9fea8593712af3c 446B / 446B 1.2s
=> => extracting sha256:38a8310d387e375e0ec6fabe047a9149e8eb214073db9f461fee6251fd936a75 0.3s
=> => extracting sha256:65052e355180ad17acd4b61613338090c587b931124cd2946b9dcf0e945b76cb 1.9s
=> => extracting sha256:9cf3157062e6e87bb84921361434e0c930e8f73584eb06114cf7563178355253 0.1s
=> => extracting sha256:dcfb660a4e70b11e73b64689666db5db05706d6d6a9bb950d9fea8593712af3c 0.0s
=> [internal] load build context                                                0.0s
=> => transferring context: 87.91kB                                              0.0s
=> [builder 2/4] WORKDIR /app                                                  2.8s
=> [builder 3/4] COPY package*.json ./                                         0.7s
=> [builder 4/4] RUN npm ci --only=production                                 2.5s
=> [stage-1 3/8] COPY --from=builder /app/node_modules ./node_modules         1.0s
=> [stage-1 4/8] COPY --from=builder /app/package*.json ./                    0.3s
=> [stage-1 5/8] COPY server.js ./                                             0.3s
=> [stage-1 6/8] COPY public ./public/                                         0.6s
=> [stage-1 7/8] RUN addgroup -g 1001 -S nodejs && adduser -S nextjs -u 1001 1.5s
=> [stage-1 8/8] RUN chown -R nextjs:nodejs /app                             8.4s
=> exporting to image                                                         0.7s
=> => exporting layers                                                            0.6s
=> => writing image sha256:154c86de7a6e40b72526acd2c92e3b84ff6bb6e6a0604082df23f20412e47e61 0.0s
=> => naming to us.icr.io/sny-labs-harshsingh15/guestbook:v1                 0.0s
theia@theiaopenshift-harshsingh15:/home/project/guestbook/v1/guestbook$
```

6. Push the image to IBM Cloud Container Registry.

► Hint!

```
docker push us.icr.io/sny/MY_NAMESPACE/guestbook:v1
```

```
theia@theiaopenshift-harshsingh15:/home/project/guestbook/v1/guestbook$ docker push us.icr.io/$MY_NAME
SPACE/guestbook:v1
The push refers to repository [us.icr.io/sn-labs-harshsingh15/guestbook]
a6e8e691256b: Pushed
30ea9365204d: Pushed
4d9947b4b1e1: Pushed
5c011032c747: Pushed
35f03ea5add6: Pushed
03d093412961: Pushed
8968817e29d6: Pushed
9fd30ff9d315: Pushed
875b8fb676b2: Pushed
8c0ba6a3f014: Pushed
3e01818d79cd: Pushed
v1: digest: sha256:ece75e3945da9b7bd3d6c14484d281dd77aa82598730c578e7c8083ba234b013 size: 2617
theia@theiaopenshift-harshsingh15:/home/project/guestbook/v1/guestbook$
```

Note: If you have tried this lab earlier, there might be a possibility that the previous session is still persistent. In such a case, you will see a **'Layer already Exists'** message instead of the **'Pushed'** message in the above output. We recommend you to proceed with the next steps of the lab.

7. Verify that the image was pushed successfully.

```
ibmcloud cr images
```

```
theia@theiaopenshift-harshsingh15:/home/project/guestbook/v1/guestbook$ ibmcloud cr images
Listing images...

Repository                                     Tag
Size      Security status      Digest      Namespace      Created
us.icr.io/sn-labs-harshsingh15/guestbook      v1
49 MB      -                     ece75e3945da sn-labs-harshsingh15 6 minutes ago
us.icr.io/sn-labsassets/categories-watson-nlp-runtime      latest
3.1 GB      -                     6b01b1e5527b sn-labsassets      2 years ago
us.icr.io/sn-labsassets/classification-watson-nlp-runtime  latest
4.0 GB      -                     dbd407898549 sn-labsassets      2 years ago
us.icr.io/sn-labsassets/concepts-watson-nlp-runtime        latest
3.2 GB      -                     1e4741f10569 sn-labsassets      2 years ago
us.icr.io/sn-labsassets/custom-watson-nlp-runtime          latest
6.5 GB      -                     f6513e19a33d sn-labsassets      2 years ago
us.icr.io/sn-labsassets/detagg-watson-nlp-runtime          latest
2.7 GB      -                     38916c2119fc sn-labsassets      2 years ago
us.icr.io/sn-labsassets/emotion-watson-nlp-runtime         latest
1c9de1d27318 sn-labsassets      2 years ago
```

Assessment:

For Option 1: AI-Graded Submission and Evaluation: Copy and paste the terminal output and save it in a text file named `crimages` for the final project submission and evaluation.

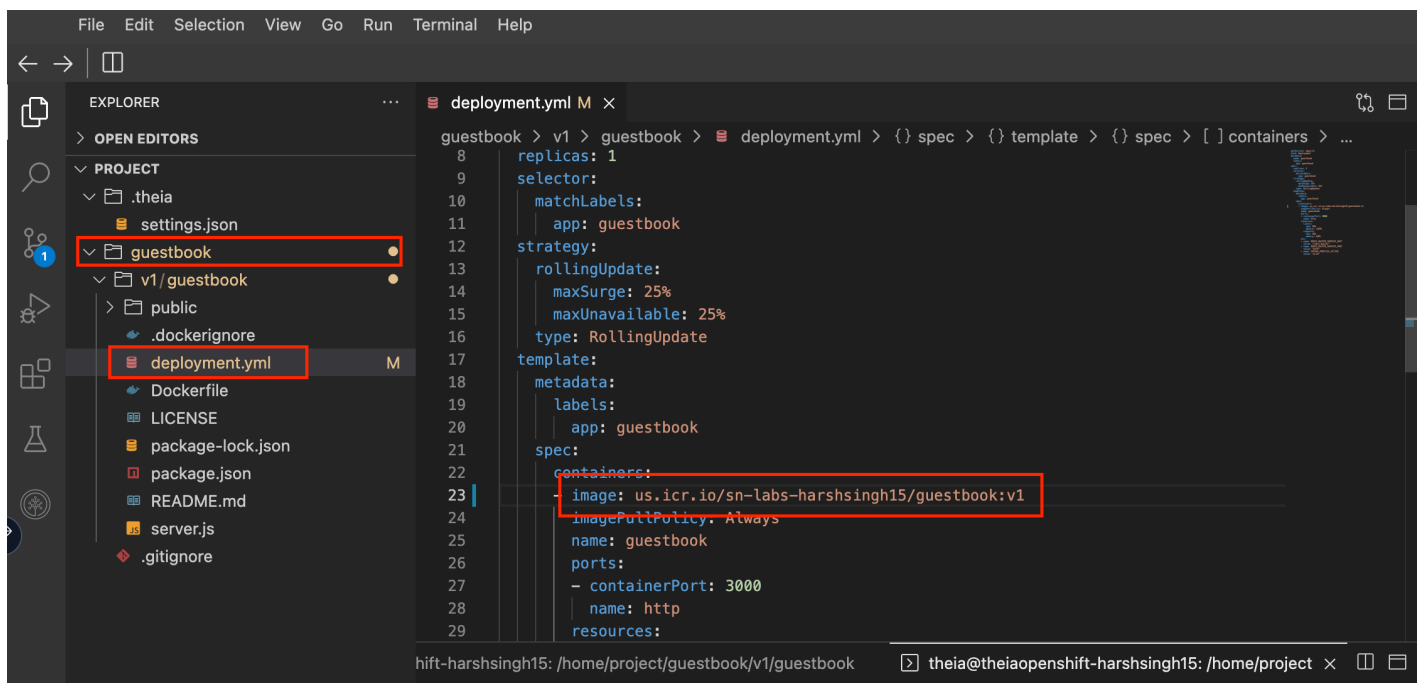
For Option 2: Peer-Graded Submission and Evaluation: Take a screenshot of the output of Step 6 and save it as `crimages.jpeg` or `crimages.png` for the Peer Assignment.

8. Open the `deployment.yml` file in the `v1/guestbook` directory and view the code for the deployment of the application:

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: guestbook
  labels:
    app: guestbook
spec:
  replicas: 1
  selector:
    matchLabels:
      app: guestbook
  strategy:
    rollingUpdate:
      maxSurge: 25%
      maxUnavailable: 25%
    type: RollingUpdate
  template:
    metadata:
      labels:
        app: guestbook
    spec:
      containers:
        - image: us.icr.io/<your SN Labs namespace>/guestbook:v1
          imagePullPolicy: Always
          name: guestbook
          ports:
            - containerPort: 3000
              name: http
          resources:
            limits:
              cpu: 50m
              requests:
                cpu: 20m
            env:
              - name: REDIS_MASTER_SERVICE_HOST
                value: 'redis-master'
              - name: REDIS_MASTER_SERVICE_PORT
                value: '6379'
              - name: SPRING_PROFILES_ACTIVE
                value: 'prod'
          restartPolicy: Always
      replicas: 1
```

Note: Replace `<your sn labs namespace>` with your SN labs namespace. To check your SN labs namespace, please run the command `ibmcloud cr namespaces`.

• It should look as below:



9. Apply the deployment using:

```
kubectl apply -f deployment.yaml
```

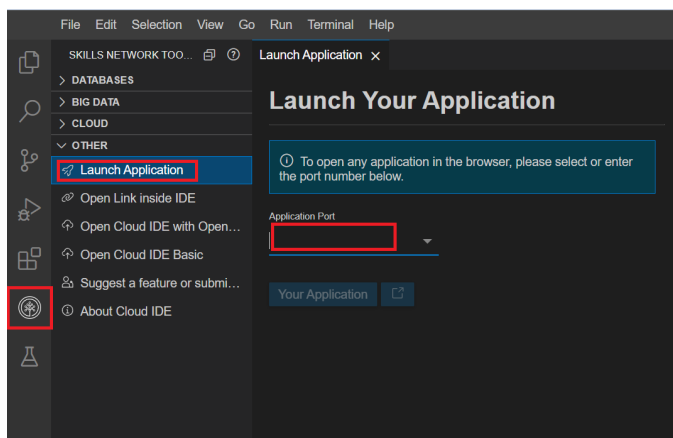
```
theia@theiaopenshift-harshsingh15:/home/project/guestbook/v1/guestbook$ kubectl apply -f deployment.yaml
deployment.apps/guestbook created
theia@theiaopenshift-harshsingh15:/home/project/guestbook/v1/guestbook$
theia@theiaopenshift-harshsingh15:/home/project/guestbook/v1/guestbook$
```

10. Open a New Terminal and enter the below command to view your application:

```
kubectl port-forward deployment.apps/guestbook 3000:3000
```

```
theia@theiaopenshift-harshsingh15:/home/project/guestbook/v1/guestbook$ kubectl port-forward deployment.apps/guestbook 3000:3000
Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::]:3000 -> 3000
Handling connection for 3000
Handling connection for 3000
Handling connection for 3000
Handling connection for 3000
Handling connection for 3000
Handling connection for 3000
Handling connection for 3000
Handling connection for 3000
Handling connection for 3000
Handling connection for 3000
Handling connection for 3000
Handling connection for 3000
```

11. Launch your application on port 3000. Click on the Skills Network button on the right, it will open the "Skills Network Toolbox." Then, click **Other**, then **Launch Application**. From there, you should be able to enter the port and launch.



12. Now you should be able to see your running application. Please copy the app URL, which will be given as below:

**Assessment:**

For Option 1: AI-Graded Submission and Evaluation: Copy and paste the code of the file index.html and save it in a text file named `app` for the final project submission and evaluation. You will be required to provide the code from this file, showing the default title and header.

For Option 2: Peer-Graded Submission and Evaluation: Take a screenshot of your deployed application and save it as `app.jpg` OR `app.png` for the Peer Assignment.

13. Try out the guestbook by putting in a few entries. You should see them appear above the input box after you hit **Submit**.

Autoscale the Guestbook application using Horizontal Pod Autoscaler

1. Autoscale the Guestbook deployment using `kubectl autoscale deployment`.

► Hint!

```
kubectl autoscale deployment guestbook --cpu-percent=5 --min=1 --max=10
```

```
theia@theiaopenshift: /home/project$ kubectl autoscale deployment guestbook --cpu-percent=5 --min=1 --max=10
horizontalpodautoscaler.autoscaling/guestbook autoscaled
```

2. You can check the current status of the newly-made HorizontalPodAutoscaler, by running:

```
kubectl get hpa guestbook
```

The current replicas is 0 as there is no load on the server.

Assessment:

For Option 1: AI-Graded Submission and Evaluation: Copy and paste the terminal output and save it in a text file named `hpa` for the final project submission and evaluation.

For Option 2: Peer-Graded Submission and Evaluation: Take a screenshot of your Horizontal Pod Autoscaler and save it as `hpa.jpg` OR `hpa.png` for the Peer Assignment.

3. Open another new terminal and enter the below command to generate load on the app to observe the autoscaling. (Please ensure your port-forward command is running. In case you have stopped your application, please run the port-forward command to re-run the application at port 3000.)

```
kubectl run -i --tty load-generator --rm --image=busybox:1.36.0 --restart=Never -- /bin/sh -c "while sleep 0.01; do wget -q -O- <your app URL>; done"
```

- Please replace your app URL in the `<your app URL>` part of the above command.

Note: Use the same copied URL which you obtained in step 11 of the previous task.

The command will be as below:

```
theia@theiaopenshift: /home/project$ kubectl run -i --tty load-generator --rm --image=busybox:1.28 --restart=Never -- /bin/sh -c "while sleep 0.01; do wget -q -O- https://3000-theiaopenshift-2-labs-prod-theiaopenshift-4-tor01.proxy.cognitiveclass.ai/; done"
```

Note: In case you get a `load-generator already exists` error, please suffix a number after `load-generator`, for example, `load-generator-1`, `load-generator-2`.

- You will keep getting an output similar as below, which will indicate the increasing load on the app:

```
<div>
<div id="guestbook-entries">
  <link href="https://afeld.github.io/emoji-css/emoji.css" rel="stylesheet">
  <p>Waiting for database connection... <i class="em em-boat"></i></p>
</div>
<div>
  <form id="guestbook-form">
    <input autocomplete="off" id="guestbook-entry-content" type="text">
    <a href="#" id="guestbook-submit">Submit</a>
  </form>
</div>
<div>
  <p>The id "guestbook-host-address"></h2></p>
  <p><a href="/env">env</a>
  <a href="/info"/>info</a></p>
</div>
<script src="jquery.min.js"></script>
<script src="script.js"></script>
</body>
</html>
```

Note: Continue further commands in the 1st terminal

4. Run the below command to observe the replicas increase in accordance with the autoscaling:

```
kubectl get hpa guestbook --watch
```

```
theia@theiaopenshift: /home/project$ kubectl get hpa guestbook --watch
NAME          REFERENCE          TARGETS      MINPODS  MAXPODS  REPLICAS  AGE
guestbook     Deployment/guestbook  unknown/5%   1         10        0          51s
```

5. Run the above command again after 5-10 minutes and you will see an increase in the number of replicas, which shows that your application has been autoscaled.

```
theia@theiaopenshift-harshsingh15:/home/project/guestbook/v1/guestbook$ kubectl get hpa guestbook --watch
NAME          REFERENCE          TARGETS      MINPODS  MAXPODS  REPLICAS  AGE
guestbook     Deployment/guestbook  10%/5%       1         10        2          114s
guestbook     Deployment/guestbook  55%/5%       1         10        2           2m1s
guestbook     Deployment/guestbook  55%/5%       1         10        4           2m16s
guestbook     Deployment/guestbook  35%/5%       1         10        8           2m31s
guestbook     Deployment/guestbook  35%/5%       1         10       10           2m46s
```

Assessment:

For Option 1: AI-Graded Submission and Evaluation: Copy and paste the terminal output of your Autoscaler details and save it in a text file named `hpa2` for the final project submission and evaluation.

For Option 2: Peer-Graded Submission and Evaluation: Take a screenshot of your Autoscaler details and save it as `hpa2.jpg` OR `hpa2.png` for the Peer Assignment.

6. Run the below command to observe the details of the horizontal pod autoscaler:

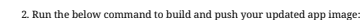
```
kubectl get hpa guestbook
```

- Please close the other terminals where load generator and port-forward commands are running

Note: Please run all the commands in the 1st terminal unless mentioned to use a new terminal.

4. Please update the title and header in `index.html` to **Guestbook - v2**

► Hint!



► Hint!

```
docker build . -t us.icr.io/$MY_NAMESPACE/guestbook:v2 && docker push us.icr.io/$MY_NAMESPACE/guestbook:v2
```

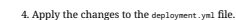


For Option 1: AI-Graded Submission and Evaluation: Copy and paste the terminal output and save it in a text file named `upgquestbook` for the final project submission and evaluation. Your saved output must include the final digest line shown after a successful push.

For Option 2: Peer-Graded Submission and Evaluation: Take a screenshot of your updated image and save it as `upquestbook.jpeg` or `upquestbook.png` for the Peer Assignment.

8. Update the value of the TAG in the `deployment.yml` to **v2**. Also Update the values of the CPU in the `deployment.yml` to `cpu: 5m` and `cpu: 2m` as below:

► Hint!



► Hint!

```
kubectl apply -f deployment.yml
```



For Option 1

For Option 2: Peer-Graded Submission and Evaluation: Take a screenshot of the details of the output of Step 4 and save it as `deployment.jpeg` OR `deployment.png` for the Peer Assignment.

5. Run the port-forward command again to start the app in new terminal:

```
kubectl port-forward deployment.apps/guestbook 3000:3000
```

6. Launch your application on port 3000. Click on the Skills Network button on the right, it will open the "Skills Network Toolbox." Then, click **Other**, then **Launch Application**. From there, you should be able to enter the port and launch.



https://wandap-3000.theiaopenshiftnext-0-labs-prod-theiaopenshift-4-tor01.proxy.cognitiveclass.ai/_env/info/hello

For Option 1: AI-Graded Submission and Evaluation: Copy and paste the code of your updated index.html and save it in a text file named `up-app` for the final project submission and evaluation. You'll be asked to paste the updated code from index.html showing the title and header as "Guestbook - v2".

Note: Please stop the application before running the next steps.

8. Run the below command to see the history of deployment rollouts:

```
kubectl rollout history deployment/guestbook
```

9. Run the below command to see the details of Revision of the deployment rollout:

```
kubectl rollout history deployments guestbook --revision=2
```

For Option 1: AI-Graded Submission and Evaluation: Copy and paste the terminal output of the details of the correct Revision and save it in a text file named `rev` for the final project submission and evaluation.

For Option 2: Peer-Graded Submission and Evaluation: Take a screenshot of the details of the correct Revision and save it as `rev.jpeg` or `rev.png` for the Peer Assignment.

10. Run the below command to get the replica sets and observe the deployment, which is being used now:

```
kubectl get rs
```

```
theia@theiaopenshift-harshsingh15:/home/project/guestbook/v1/guestbook$ kubectl get rs
NAME                                DESIRED   CURRENT   READY   AGE
guestbook-59d8bd97fb                0         0         0       21m
guestbook-694dbb8c4c                1         1         1       3m51s
openshift-web-console-65fcb9c795    2         2         2       25m
```

11. Run the below command to undo the deployment and set it to Revision 1:

```
kubectl rollout undo deployment/guestbook --to-revision=1
```

```
theia@theiaopenshift-harshsingh15:/home/project/guestbook/v1/guestbook$ kubectl rollout undo deployment/guestbook --to-revision=1
deployment.apps/guestbook rolled back
```

12. Run the below command to get the replica sets after the rollout has been undone. The deployment being used would have changed as below:

```
kubectl get rs
```

```
theia@theiaopenshift-harshsingh15:/home/project/guestbook/v1/guestbook$ kubectl get rs
NAME                                DESIRED   CURRENT   READY   AGE
guestbook-59d8bd97fb                1         1         1       23m
guestbook-b94dbb8c4c                0         0         0       5m59s
openshift-web-console-65fcb9c795    2         2         2       27m
```

Assessment:

For Option 1: AI-Graded Submission and Evaluation: Copy and paste the terminal output and save it in a text file named `rs` for the final project submission and evaluation.

For Option 2: Peer-Graded Submission and Evaluation: Take a screenshot of the output of Step 12 and save it as `rs.jpeg` or `rs.png` for the Peer Assignment.

Note: The rollout may take a few minutes to stabilize. Please wait until the replica set matches the state shown in the screenshot above before copying the output for future reference.

Congratulations! You have completed the final project for this course. Do not log out of the lab environment (you can close the browser though) or delete any of the artifacts created during the lab, as these will be needed for the next lab,Optional: Deploy Guestbook App from the OpenShift Internal Registry.

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